

Sensor-Set Conditioning for Prognostic Posterior Consistency: Gains Under Operational Variability and the Importance-Awareness Boundary

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October 3, 2026

Abstract

Deep prognostic models typically assume a fixed sensor configuration, yet deployed systems routinely lose sensors to outages, failures, and maintenance. We study whether explicitly conditioning remaining useful life (RUL) predictions on the available sensor set improves cross-view posterior consistency—the requirement that predictions for the same asset viewed through different sensor subsets remain geometrically compatible. In a preregistered evaluation on a controlled synthetic benchmark with exact oracle posteriors (96 assets; independent test and lockbox splits), sensor-set conditioning reduces cross-view posterior inconsistency by 38.6% (90% CI: 34.8–42.4%) relative to observation-agnostic baselines when alternative views remain task-sufficient, and the gain persists under a four-fold reduction in Monte Carlo sampling budget. Under permanent loss of critical sensors the advantage disappears: exploratory analysis indicates that the architecture detects information quantity more readily than functional sensor importance, while a standard baseline shows the opposite pattern. On C-MAPSS, a shared-model evaluation reduces distance to the full view but reverses for two equally sized reduced views, while point MAE remains close to the baseline. This conditional pattern sharpens the boundary of observation-process conditioning.

Keywords: prognostics and health management; remaining useful life prediction; variable sensor configurations; missing sensor data; probabilistic forecasting; uncertainty quantification; posterior consistency; calibration

1 Introduction

1.1 Motivation: The Observation Process Problem

Industrial prognostic systems operate in environments where sensor configurations rarely remain static. Communication dropouts create intermittent data gaps. Individual sensors fail and await replacement. Maintenance interventions temporarily disable portions of the sensing infrastructure. During these events, predictive maintenance models must continue to provide reliable remaining useful life (RUL) estimates, yet most deep learning architectures assume the observation process—the subset and quality of available sensors—remains fixed throughout deployment.

This mismatch creates a subtle but consequential failure mode. A model may achieve acceptable pointwise prediction accuracy while producing *incompatible posterior distributions* when the same asset is viewed through different observation configurations. For example, predictions made with a full sensor complement may yield narrow, confident distributions, while predictions from a reduced sensor set produce wider distributions that do not properly overlap the full-sensor

posterior—even though both views ostensibly describe the same underlying degradation state. Such geometric inconsistencies undermine uncertainty-aware decision making, where maintenance scheduling depends not just on the posterior mean but on tail probabilities, prediction intervals, and risk assessments.

1.2 Prior Work and the Unresolved Question

The methodological foundation separates the observation mechanism from the latent quantity being predicted [1, 2]. Neural imputation methods such as BRITS and CSDI [3, 4], permutation-invariant set encoders and attention mechanisms [5, 6, 7], and conditional neural processes [8] each provide a way to represent incomplete or changing inputs. For probabilistic prognostics, proper scoring rules and calibration diagnostics provide complementary checks on forecast quality [9, 10, 11]. These foundations motivate the observation-process perspective, while the RUL literature below supplies the closest application-specific comparisons.

Recent prognostics research has begun to address variable and degraded sensing, but its primary objectives differ across several neighboring lines of work. Methods for missing or faulty sensors typically aim to preserve point-prediction accuracy by recovering the input or learning a representation that is robust to incomplete channels. Niu et al. [12] use real-time data imputation within a dynamic adaptive prognostics pipeline, while Liu et al. [13] jointly optimize sensor reconstruction and RUL prediction with a spatio-temporal multi-task network. Zhang et al. [14] introduce a variable-sensor scenario and use cross-channel attention and data augmentation to stabilize RUL prediction. Wang et al. [15] model partially observed sensor streams with a graph feature-gated graph attention network, thereby making relations among sensor channels explicit. These studies establish strong precedents for learning under changing sensor availability, yet their main comparisons remain single-view prediction errors against the observed RUL target.

A second line of work focuses on uncertainty quantification and reliability. Sankararaman [16] frames prognostic uncertainty as an uncertainty propagation problem, and Li et al. [17] distinguish epistemic and aleatoric contributions in a Bayesian deep-learning RUL framework. More recent work combines missing-value modeling with distribution-free interval construction: Wang et al. [18] use diffusion-based imputation, metric learning, and conformal prediction to obtain statistically valid intervals under randomly missing or partially faulty sensors. Xu et al. [19] further combine Bayesian predictive distributions with isotonic uncertainty calibration and split conformal intervals for RUL prediction. The recent review by Lin et al. [20] places these approaches within a broader taxonomy of statistical, learning-based, and hybrid uncertainty analysis. Related work on conformal RUL intervals and domain generalization [21, 22, 23, 24] likewise emphasizes coverage, robustness, or transfer across operating conditions.

Sensor degradation extends the same concern beyond binary availability. Earlier reliability studies modelled joint system–sensor degradation in RUL inference using state-space and filtering methods [25, 26]. Zhao et al. [27] extend this line to rolling bearings by jointly modelling bearing and sensor degradation so that measurement quality can be updated during RUL inference. This perspective is useful for defining the observation process more broadly: availability, measurement quality, and functional sensor importance are related but distinct aspects of what a predictor observes.

These neighboring methods answer important questions about recovering missing measurements, maintaining point accuracy, constructing valid intervals, or exploiting sensor relations. However, they do not make the paired geometric relationship between predictive distributions from different sensor views of the same asset their primary estimand. The question studied here is different: for the same asset and query time, when two sensor sets remain task-sufficient, do they induce posterior distributions with a coherent geometric relationship? We therefore treat the observation

process as an explicit conditioning variable and make cross-view posterior inconsistency the primary estimand. This distinction also explains the role of the boundary analysis: a model may recognize that a sensor set has changed while still failing to represent the functional importance of the missing channels.

Time-series foundation models have broadened the forecasting model family and motivated more realistic transfer analyses [28, 29, 30]. They provide useful modeling backbones, but do not by themselves define a paired probabilistic estimand for changing sensor sets. Our study isolates that observation-process question within a controlled prognostic setting.

1.3 Our Approach: Explicit Observation-Process Conditioning

We propose an architecture that treats the observation process as a *first-class conditioning variable*, analogous to how conditional generative models explicitly condition on class labels or attributes. The key insight is that posterior geometry should *explicitly* depend on which sensors are available, not merely adapt implicitly through input masking.

Concretely, our model—termed Prognostic Observation-Process-Directed (PROP-DIRECT)—embeds the sensor set configuration into a latent representation, then conditions all posterior computations on this representation alongside the observed sensor values. This design provides three potential benefits:

1. **Explicit uncertainty propagation:** The model learns to widen posteriors appropriately when critical sensors are missing, rather than maintaining false confidence.
2. **Cross-view consistency:** Predictions from different observation views of the same asset should remain geometrically compatible—overlapping posteriors for overlapping information.
3. **Graceful degradation:** As sensor availability decreases, posterior uncertainty should increase smoothly rather than exhibiting discontinuous failures.

1.4 Scope and Contributions

We test this question with a preregistered protocol that separates model development from independent test and lockbox evaluations. The central hypothesis is conditional: conditioning should help when alternative sensor views remain *task-sufficient*, meaning that the available measurements still support reliable RUL prediction. The protocol defines task sufficiency through oracle-posterior quality thresholds (Section 2.4) and then examines what happens when functionally critical sensors are removed.

Our contributions are:

1. **Empirical result under task sufficiency:** Across 96 synthetic assets with controlled observation processes, sensor-set conditioning reduces cross-view posterior inconsistency by 38.6% (90% CI: 34.8–42.4%) relative to observation-agnostic baselines. A complementary shared-model analysis on the public C-MAPSS simulated benchmark reveals that the geometry gain depends on which view pair is compared (Section 3.5).
2. **Boundary characterization under information loss:** When critical sensors are removed, the architecture does not maintain its advantage across three pre-registered superiority criteria, locating the boundary of the conditioning mechanism.

3. **Mechanistic insight via exploratory analysis:** Post-hoc correlation analysis reveals *why* the boundary exists: the model successfully detects magnitude of information loss (oracle-model correlation $r > 0.53$), but shows a 9.9% weaker correlation when importance matters. Because prediction cases share assets and query structure, this contrast is treated as exploratory rather than as a confirmatory independent-sample test. Crucially, the pattern is architecture-specific—a standard baseline shows the opposite direction.
4. **Quality-geometry separation:** In the synthetic information-loss evaluation, predictive quality remains competitive even when geometric consistency degrades, separating posterior shape from point prediction accuracy.

1.5 Study Design and Scope

We use a synthetic generator because exact oracle posteriors and controlled observation processes make the geometric comparison identifiable. A complementary analysis on the public C-MAPSS simulated turbofan benchmark asks whether the scoring pattern extends beyond this generator; the Discussion places that result alongside the still-open question of naturally occurring sensor loss.

1.6 Paper Organization

Section 2 describes the PROP-DIRECT architecture, study protocol, metrics, and benchmark design. Section 3 reports the primary analysis under sufficient observation, the information-loss boundary, the exploratory mechanism analysis, predictive quality and calibration, and the complementary C-MAPSS evaluation. Section 4 interprets these findings, positions them relative to existing approaches, and provides deployment guidance. Section 5 summarizes contributions and outlines future directions.

2 Methods

2.1 The PROP-DIRECT Architecture

2.1.1 Motivation and Design Principles

Standard prognostic models treat sensor availability as fixed: the architecture assumes a specific set of sensors will be present at inference time. When sensors become unavailable, common approaches either impute missing values or use input masking, but neither explicitly represents *which observation process is active*. Our architecture addresses this gap by treating the observation process as a first-class conditioning variable, analogous to how conditional generative models explicitly condition on class labels.

2.1.2 Architecture Overview

PROP-DIRECT (Prognostic Observation-Process-Directed) consists of three main components:

Figure 1 summarizes the information flow and makes the distinction from simple input masking explicit: the sensor-set embedding is injected into every recurrent step before the posterior is formed.

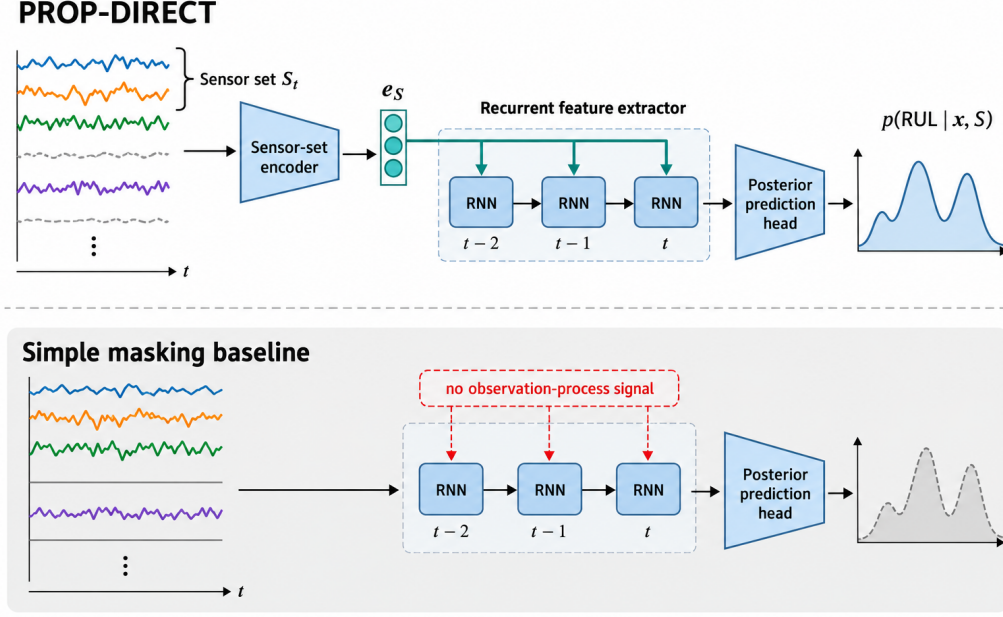


Figure 1: Observation-process conditioning in PROP-DIRECT. The model encodes the available sensor set as $\mathbf{e}_S$ and injects that representation into each recurrent step before predicting the RUL posterior. The lower strip shows the masking baseline, which replaces missing values but provides no explicit observation-process signal.

Sensor-Set Encoder Given an observation at time t from asset a , let S_t denote the set of available sensors and $\mathbf{x}_t \in \mathbb{R}^{|S_t| \times d}$ the observed sensor values over a historical window of length d . The sensor-set encoder maps S_t to a fixed-dimensional embedding $\mathbf{e}_S \in \mathbb{R}^k$:

$$\mathbf{e}_S = f_{\text{enc}}(S_t; \theta_{\text{enc}})$$

We implement f_{enc} as a set encoder using a learned lookup table: each sensor has an embedding vector, and $\mathbf{e}_S$ is computed as the element-wise mean of embeddings for sensors in S_t . This permutation-invariant design ensures the representation does not depend on sensor ordering.

Observation-Conditioned Feature Extraction The observed sensor values $\mathbf{x}_t$ are processed by a recurrent encoder (LSTM or GRU) that is *conditioned* on $\mathbf{e}_S$:

$$\mathbf{h}_t = f_{\text{rnn}}(\mathbf{x}_t, \mathbf{e}_S; \theta_{\text{rnn}})$$

Conditioning is implemented by concatenating $\mathbf{e}_S$ with the hidden state at each recurrent step, enabling the feature extraction process to adapt based on sensor availability. This differs from standard masking approaches where the recurrent network sees only the values (or masked versions) without explicit knowledge of the observation configuration.

Posterior Prediction Head The extracted features $\mathbf{h}_t$ are passed to a prediction head that outputs distributional parameters for the remaining useful life (RUL) posterior. We model RUL as a mixture of log-normal distributions to capture multi-modal degradation behavior:

$$p(\text{RUL} | \mathbf{x}_t, S_t) = \sum_{i=1}^M \pi_i(\mathbf{h}_t, \mathbf{e}_S) \cdot \text{LogNormal}(\mu_i(\mathbf{h}_t, \mathbf{e}_S), \sigma_i(\mathbf{h}_t, \mathbf{e}_S))$$

Critically, all distributional parameters $\{\pi_i, \mu_i, \sigma_i\}$ depend on *both* the observed data $\mathbf{h}_t$ and the sensor-set embedding $\mathbf{e}_S$. This dual conditioning enables the posterior geometry to explicitly vary with observation configuration.

2.1.3 Training Objective

The model is trained to maximize the log-likelihood of observed failure times across diverse sensor configurations:

$$\mathcal{L} = \sum_{(a,t,S) \in \mathcal{D}} \log p(\text{RUL}_{a,t} \mid \mathbf{x}_{a,t}, S; \theta)$$

where $\mathcal{D}$ is the training dataset covering multiple assets, query times, and sensor configurations. The training data is deliberately constructed to include a variety of observation processes (not just the full sensor complement) so the model learns to adjust posteriors appropriately for different configurations.

2.2 Baseline Architectures

To isolate the contribution of observation-process conditioning, we compare PROP-DIRECT to six baseline architectures, all trained on the same data and evaluated with the same protocol:

B-STDPROB (Standard Probabilistic Baseline) A recurrent neural network with a mixture-of-log-normals output distribution, but *without* sensor-set conditioning. The architecture sees masked inputs (zeros for missing sensors) but does not receive explicit information about which sensors are available. This is the primary comparison model for isolating the effect of conditioning.

B-INDEP (Independent Sensor Baseline) Treats each sensor independently without modeling temporal or cross-sensor dependencies. Provides a lower bound on performance.

B-INVAR (Time-Invariant Baseline) Uses only static asset features without temporal modeling. Tests whether observation-process conditioning provides value beyond time-invariant prediction.

B-CDE (Conditional Density Estimation) A density estimation approach that models $p(\text{RUL} \mid \mathbf{x})$ directly via kernel methods rather than parametric distributions. Provides an alternative uncertainty quantification strategy.

B-SET (Set-Based Aggregation) Aggregates sensor values using permutation-invariant pooling (max, mean, attention-weighted) without explicit recurrence. Tests whether set structure alone suffices without temporal modeling.

B-WIDEN (Wider Network) A standard architecture (like B-STDPROB) but with increased capacity (more layers, wider hidden states). Controls for the possibility that PROP-DIRECT’s gains come from additional parameters rather than conditioning structure.

All models are trained with the same optimization procedure (Adam optimizer, learning rate schedule, early stopping criteria) and hyperparameters are selected via validation set performance to ensure fair comparison.

2.3 Synthetic Data Generator

2.3.1 Rationale for Synthetic Data

Rigorous evaluation of observation-process conditioning requires knowing the *true* posterior distribution under different sensor configurations—what we term the “oracle posterior.” Real-world PHM datasets do not provide this ground truth, making it impossible to directly measure whether predicted posteriors are geometrically consistent with true posteriors.

We therefore employ a controlled synthetic generator based on established degradation physics. This generator enables:

- **Oracle posterior computation:** Exact calculation of $p(\text{RUL} \mid \text{observations, degradation state})$ for any sensor configuration.
- **Causal manipulation:** Controlled variation of observation processes while holding asset state fixed, enabling causal attribution.
- **Importance ground truth:** Explicit specification of which sensors are critical versus redundant, necessary for evaluating importance-awareness.

We emphasize that synthetic data limits external validity—findings may not transfer to real industrial assets—but it is essential for the mechanistic analysis conducted here. Section 4.4 discusses this trade-off in detail.

2.3.2 Generator Structure

The generator models asset degradation as a latent Markov process with observable sensor readings:

Degradation Dynamics Each asset’s health state $\mathbf{z}_t \in \mathbb{R}^D$ evolves according to:

$$\mathbf{z}_{t+1} = g(\mathbf{z}_t, \boldsymbol{\theta}_{\text{phys}}) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \Sigma_{\text{proc}})$$

where $g(\cdot)$ is a physics-based degradation function (e.g., exponential bearing wear, cumulative fatigue damage) parameterized by $\boldsymbol{\theta}_{\text{phys}}$, and $\boldsymbol{\epsilon}_t$ represents process noise. The asset fails when any component of $\mathbf{z}_t$ crosses a failure threshold, defining the true RUL.

Observation Process Sensor readings $\mathbf{x}_t$ are noisy functions of the latent state:

$$x_{s,t} = h_s(\mathbf{z}_t) + \eta_{s,t}, \quad \eta_{s,t} \sim \mathcal{N}(0, \sigma_s^2)$$

where $h_s(\cdot)$ maps latent health to sensor s ’s measurement and $\eta_{s,t}$ is observation noise. Different sensors have different mappings h_s :

- **Critical sensors** directly observe components near failure thresholds (e.g., vibration monitoring bearing wear).

- **Redundant sensors** observe the same physical quantity with independent noise (e.g., multiple temperature probes on the same subsystem).
- **Ancillary sensors** observe tangentially related quantities (e.g., ambient temperature when bearing temperature is critical).

Observation Configurations The generator samples sensor availability configurations according to specified distributions:

- **Full configuration:** All sensors available (baseline).
- **Sufficient deletions:** Remove ancillary or redundant sensors, maintaining critical coverage.
- **Critical deletions:** Remove sensors that directly observe near-failure states.
- **Random deletions:** Remove sensors uniformly at random, matched in count to critical deletions.

The causal structure of this generator is shown in Figure 2. It separates the latent degradation state from noisy measurements and makes the critical, redundant, and ancillary sensor roles explicit.

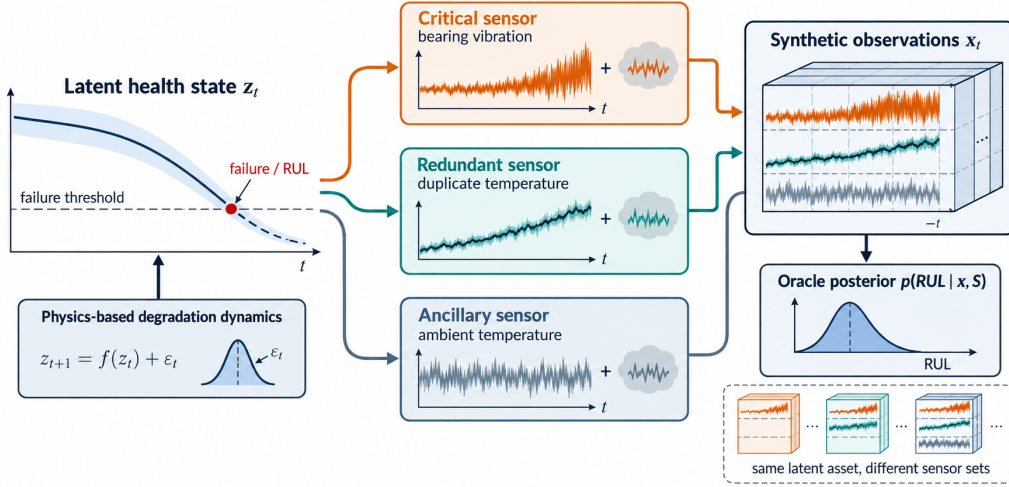


Figure 2: Synthetic degradation-data generator. A latent health state evolves under physics-based dynamics and crosses a failure threshold at the true RUL. Noisy observation channels produce critical, redundant, and ancillary sensors, while different sensor subsets of the same latent asset support exact oracle-posterior computation.

2.3.3 Asset Population

We generate 96 synthetic assets with the following variation:

- **Degradation parameters θ_{phys} :** Sampled to produce heterogeneous failure times (mean RUL: 5000 cycles, range: 2000–8000).
- **Observation noise σ_s^2 :** Varies across sensors and assets to simulate sensor quality differences.

- **Operating conditions:** Three regimes (low/medium/high load) affecting degradation rates.
- **Sensor topologies:** Different mappings $h_s(\cdot)$ to simulate domain variety (e.g., vibration-dominated vs. temperature-dominated degradation).

Each asset is observed from initial deployment until failure, with queries sampled at regular intervals (every 50 cycles) throughout its lifecycle. This produces approximately $96 \text{ assets} \times 100 \text{ query times} \times \text{multiple observation configurations} = \text{tens of thousands of evaluation cases}$.

2.4 Study Design and Evaluation Protocol

2.4.1 Preregistration and Data Partitioning

To ensure statistical integrity, we employ a preregistered evaluation protocol where all decision criteria, thresholds, and analysis plans are specified *before* accessing evaluation outcomes. The protocol proceeds in stages:

Figure 3 provides a compact view of the four stages and the separation between exploratory development, test replication, and the lockbox used for the primary claim.

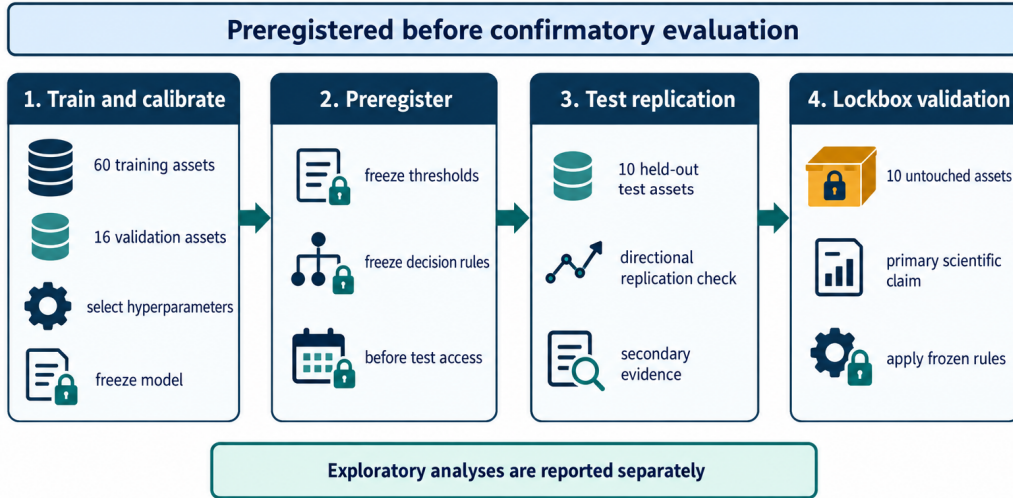


Figure 3: Preregistered evaluation protocol. Model development and threshold selection are completed before held-out evaluation. The test split serves as a directional replication check, whereas the untouched lockbox is evaluated with frozen decision rules and supplies the primary scientific claim.

Stage 1: Model Development and Freezing

1. Train all seven models (PROP-DIRECT + six baselines) on a designated training set (60 assets).
2. Select hyperparameters via validation set performance (16 assets).
3. Freeze model checkpoints and record them in a versioned registry.

Stage 2: Threshold Specification

1. Using the validation set and power analysis, specify:
 - Non-inferiority margins for quality metrics (MAE, CRPS).
 - Superiority margins for geometric consistency metrics.
 - Acceptability thresholds for absolute geometry.
2. Commit these thresholds to a frozen registry before scoring test or lockbox data.

Stage 3: Test Split Evaluation

1. Score all models on a held-out test split (10 assets) using frozen checkpoints.
2. Compare outcomes to preregistered criteria.
3. This stage serves as a “replication check”—we expect consistent direction of effects with the validation set but do not make final scientific claims based on test results alone.

Stage 4: Lockbox Validation

1. Score all models on a fully held-out lockbox split (10 assets) that no human has inspected during development.
2. Apply preregistered decision rules to lockbox outcomes.
3. Lockbox results are the primary basis for scientific claims.

This staging records which decisions are made before each held-out evaluation and keeps exploratory analyses separate from the primary claim.

2.4.2 Data Split Composition

- **Training:** 60 assets, diverse degradation parameters and sensor configurations.
- **Validation:** 16 assets, used for hyperparameter selection and threshold calibration.
- **Test:** 10 assets, first held-out evaluation for replication check.
- **Lockbox:** 10 assets, fully independent validation for primary claims.

All splits are stratified by degradation regime (low/medium/high load) and sensor topology to ensure balanced representation.

The partition above applies to the sufficient-observation analysis, whose primary claims rest on the lockbox split. The information-loss analysis uses a separate frozen evaluation covering all 96 assets; its descriptive quality metrics (Table 4) therefore summarize that population rather than the 10-asset test split. Model checkpoints and decision rules were frozen before either evaluation was accessed.

2.4.3 Information-loss boundary design

The boundary analysis compares sensor deletions that differ in functional importance while holding the number of removed channels fixed. We use three pair classes:

- **Critical deletion:** sensors identified by the oracle analysis as critical for RUL prediction;
- **Random deletion:** the same number of sensors selected at random, providing a matched control for information quantity;
- **Redundant deletion:** sensors carrying overlapping information, providing a low-impact comparison.

The corresponding estimand asks whether an importance-aware architecture keeps the critical-versus-random posterior contrast closer to the oracle-defined contrast than a model that only observes masked values. This design separates the effect of how many sensors are removed from the effect of which sensors are removed.

2.5 Evaluation Metrics

2.5.1 Geometric Consistency Metrics

The primary evaluation question is whether PROP-DIRECT produces geometrically consistent posteriors across observation views of the same asset. We measure this via the *cross-view posterior inconsistency* metric:

Paired Observation Setup For each asset a at query time t , we generate predictions under two observation configurations $\mathcal{S}_1$ and $\mathcal{S}_2$ that should provide compatible information (e.g., full sensors vs. ancillary sensors removed). Denote the predicted posteriors as $p_1 = p(\text{RUL} \mid \mathbf{x}_{a,t}, \mathcal{S}_1)$ and $p_2 = p(\text{RUL} \mid \mathbf{x}_{a,t}, \mathcal{S}_2)$.

Energy Distance We quantify inconsistency using the energy distance, a proper metric on probability distributions:

$$d_E(p_1, p_2) = 2\mathbb{E}_{Y_1 \sim p_1, Y_2 \sim p_2}[|Y_1 - Y_2|] - \mathbb{E}_{Y_1, Y'_1 \sim p_1}[|Y_1 - Y'_1|] - \mathbb{E}_{Y_2, Y'_2 \sim p_2}[|Y_2 - Y'_2|]$$

This metric:

- Equals zero if and only if $p_1 = p_2$ (distributional consistency across views).
- Increases with both location shift and shape differences.
- Does not require parametric assumptions about posterior forms.

A small d_E is a consistency statement, not an accuracy statement: a model that emits the same posterior for every sensor set can attain $d_E = 0$ while ignoring the observation process. To evaluate whether the distance has the right magnitude, we therefore report the secondary oracle-relative geometry error

$$E_{\text{geometry}} = |d_E(p_{\text{model},1}, p_{\text{model},2}) - d_E(p_{\text{oracle},1}, p_{\text{oracle},2})|.$$

Lower values indicate closer agreement with the oracle-defined geometry. This post-hoc quantity is reported as an exploratory diagnostic and does not replace the preregistered cross-view consistency endpoint.

In practice, we estimate d_E via Monte Carlo sampling: draw $N = 256$ samples from each posterior (or $N = 64$ for computational budget sensitivity), compute pairwise distances, and average.

Metric Scale and Interpretation All metrics (energy distance, MAE, CRPS) are computed on *normalized RUL values*. Normalization is performed by dividing each asset’s RUL by its maximum observed lifetime. For the synthetic generator used here, the mean lifetime is 5000 cycles (range: 2000–8000), so an energy distance of 2.86×10^{-4} in normalized units corresponds to approximately 1.4 cycles of absolute distribution separation when scaled back to the raw RUL scale. Similarly, a normalized MAE of 0.015 corresponds to approximately 75 cycles of prediction error at mean scale.

This normalization ensures metrics are comparable across assets with heterogeneous lifetimes and enables threshold specification in scale-invariant units. All thresholds (τ for superiority margins, acceptability bounds for absolute geometry) are specified in these normalized units and frozen prior to evaluation.

Off-Diagonal Correction To avoid spurious correlation from comparing a sample to itself, we use an off-diagonal estimator: when computing cross-distribution expectations $\mathbb{E}_{Y_1, Y_2}[|Y_1 - Y_2|]$, we exclude pairs where the same random number stream was used. This correction is critical for the mechanistic analysis where sample efficiency matters.

Aggregation For each model and observation-process pair type (e.g., full vs. ancillary-removed), we compute:

1. Asset-level geometry: Mean energy distance over all query times for that asset.
2. Population-level geometry: Mean over all assets.

The primary estimand is the *relative reduction*:

$$\text{Relative reduction} = \frac{d_E(\text{B-INDEP}) - d_E(\text{PROP-DIRECT})}{d_E(\text{B-INDEP})} \times 100\%$$

where B-INDEP serves as the observation-agnostic baseline. A positive value indicates PROP-DIRECT is more consistent.

Baseline Selection B-INDEP is the preregistered primary comparator because it treats sensor channels independently and provides an observation-agnostic reference. We also report B-STDPROB, which shares the observation interface of PROP-DIRECT. Their near-identical energy distances on the test and lockbox splits (Table 2) provide a complementary baseline check.

2.5.2 Quality Metrics

To verify that geometric improvements do not come at the cost of prediction quality, we evaluate:

Mean Absolute Error (MAE) For RUL point predictions (posterior median):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\text{RUL}_{\text{true},i} - \text{RUL}_{\text{pred},i}|$$

Computed separately for all time points (state MAE) and time points within 500 cycles of failure (failure MAE), as accuracy near failure is often more operationally critical.

Continuous Ranked Probability Score (CRPS) A proper scoring rule for probabilistic forecasts:

$$\text{CRPS}(p, y) = \mathbb{E}_{Y \sim p}[|Y - y|] - \frac{1}{2} \mathbb{E}_{Y, Y' \sim p}[|Y - Y'|]$$

where p is the predicted distribution and y is the observed RUL. CRPS jointly evaluates calibration (whether prediction intervals cover the true value at nominal rates) and sharpness (whether intervals are tight). Lower CRPS indicates better probabilistic prediction.

Calibration diagnostics As a complementary descriptive check, we compute central-interval coverage at nominal levels of 50%, 80%, and 90% from the persisted posterior samples. We also compute the probability integral transform (PIT) for each coordinate using the mid-rank of the observed target among the predictive samples. Coverage is summarized over the three coordinates, while PIT histograms are compared with the uniform benchmark. These diagnostics are evaluated after the preregistered scoring and do not change any decision threshold or model designation.

2.5.3 Statistical Inference

Bootstrap Confidence Intervals All primary estimates are accompanied by 90% bootstrap confidence intervals computed as follows:

1. Resample assets with replacement (preserving within-asset dependencies).
2. Recompute the estimand (e.g., relative reduction) on the resampled data.
3. Repeat 2,000 times.
4. Report the 5th and 95th percentiles of the bootstrap distribution.

Decision Rules Preregistered decision criteria:

- **Superiority claim:** The 90% CI must exclude zero *and* exclude a prespecified margin τ representing the minimum scientifically meaningful effect.
- **Non-inferiority claim:** The 90% CI lower bound must exceed a non-inferiority margin (e.g., $-\tau$).
- **Quality guardrails:** CRPS and MAE must satisfy non-inferiority vs. baseline to ensure geometric gains do not degrade prediction quality.

2.6 C-MAPSS Shared-Model Evaluation

We use C-MAPSS FD001 as a complementary public simulated benchmark. The dataset contains 100 training and 100 test engine trajectories and 14 non-constant sensor channels after preprocessing. It provides a common reference for the geometry analysis while remaining distinct from field data.

We retain the full 14-channel view and construct two reduced views by deleting three channels from each; the deletion sets differ by six channels. For each method, a single predictor is trained on the union of all three views. The 100 training engines are split by engine into 80 fitting and 20 validation engines, so views of the same engine never cross a split. Unlike the synthetic experiments, which use an explicitly distributional mixture-of-log-normals head, C-MAPSS uses an MSE point-prediction head. We construct predictive samples with 128 Monte Carlo dropout passes [31]. This is a complementary uncertainty representation, not a direct replication of the synthetic posterior model. We evaluate the final window of each test engine, report energy distances on the raw RUL scale, and use engine-level bootstrap resampling for confidence intervals. Single-view MAE is reported as a point-accuracy check alongside the geometry results.

This protocol tests whether the geometry pattern survives a shared-model training setup. Because the benchmark is simulated and the reduced views are created by artificial deletion, its results are interpreted as complementary benchmark evidence rather than field validation.

2.7 Exploratory Mechanistic Analysis

Beyond the preregistered analysis, we use an exploratory oracle comparison to interpret the different regimes. It is hypothesis-generating rather than a confirmatory test.

Oracle-Model Correlation For each paired observation case, we compute:

- **Oracle distance:** The true geometric inconsistency $d_E(p_{\text{oracle},1}, p_{\text{oracle},2})$ computed from the known ground-truth degradation state.
- **Oracle-relative geometry error:** The residual $E_{\text{geometry}} = |d_E(p_{\text{model},1}, p_{\text{model},2}) - d_E(p_{\text{oracle},1}, p_{\text{oracle},2})|$.

If the model successfully tracks task difficulty, these should correlate: cases where the oracle shows large inconsistency (observation processes genuinely differ) should also exhibit large model inconsistency.

We compute Pearson correlation r separately for different observation-process pair types (critical vs. full, critical vs. random, etc.). Because prediction cases share assets and query structure, the correlations describe an effect-size pattern rather than a confirmatory independent-sample test. Asset-level or mixed-effects inference is a natural follow-up.

2.8 Computational Environment

All experiments are conducted on NVIDIA A40 GPUs with 48GB memory. Model training uses mixed-precision (FP16) arithmetic via PyTorch’s automatic mixed precision. Evaluation scoring uses full FP32 precision to ensure numerical stability in bootstrap resampling. Monte Carlo sampling for energy distance computation is parallelized across query times and assets. Total computational budget: approximately 500 GPU-hours for training all models, 100 GPU-hours for evaluation scoring across all splits and configurations.

Table 1: Resolution sensitivity on the lockbox split. Intervals are two-sided 90% bootstrap confidence intervals with 2,000 replicates.

Sampling budget	PROP-DIRECT geometry	Relative reduction	90% CI
256 (primary)	0.0002864	38.64%	[34.78%, 42.44%]
64 (check)	0.0002849	38.97%	[35.03%, 42.66%]

Table 2: Geometric consistency comparison across baselines. Energy distance for PROP-DIRECT and two observation-agnostic baselines under the sufficient observation regime. Relative reduction computed as $(d_E(\text{baseline}) - d_E(\text{PROP}))/d_E(\text{baseline}) \times 100\%$. Lower energy distance indicates better cross-view posterior consistency. Both baselines yield comparable absolute energy distance, confirming that relative improvements are robust to baseline choice.

Split	Model	Energy Distance	Relative Reduction
Test	PROP-DIRECT	2.89×10^{-4}	—
	B-STDPROB	4.57×10^{-4}	36.8%
	B-INDEP	4.66×10^{-4}	38.0%
Lockbox	PROP-DIRECT	2.86×10^{-4}	—
	B-STDPROB	4.64×10^{-4}	38.3%
	B-INDEP	4.67×10^{-4}	38.6%

Bootstrap 90% confidence intervals for the primary comparison (PROP-DIRECT vs B-INDEP): Test [35.1%, 40.7%], Lockbox [34.8%, 42.4%]. B-STDPROB and B-INDEP differ by less than 1% in absolute energy distance, indicating that the choice of observation-agnostic baseline does not materially affect conclusions.

Code and model checkpoints are archived with version control (Git) to enable exact replication. The complete evaluation pipeline, including bootstrap resampling and decision logic, is frozen prior to lockbox evaluation and committed to the archive.

3 Results

We first report the primary cross-view result under task-sufficient observation. We then locate its boundary under critical sensor loss, interpret that boundary with an exploratory oracle comparison, and separate geometric behavior from predictive quality and calibration. The section closes with a complementary shared-model evaluation on C-MAPSS. Unless stated otherwise, intervals in the synthetic analyses use 2,000 asset-level bootstrap replicates.

3.1 Cross-View Consistency Under Sufficient Observation

3.1.1 Primary Finding

When observation processes maintain task sufficiency—that is, when different sensor subsets all support reliable RUL prediction—PROP-DIRECT substantially reduces posterior inconsistency across views of the same asset. Table 1 reports the formal lockbox validation results.

The architecture achieves an absolute geometry metric of 2.86×10^{-4} on the lockbox split, well below the pre-specified acceptability threshold of 3.5×10^{-4} . More importantly, this represents a **38.6% reduction** (90% CI: 34.8–42.4%) in cross-view inconsistency relative to the observation-agnostic baseline (B-INDEP). The test split yields a consistent 38.1% reduction

(90% CI: 34.2–41.9%), confirming the effect is not an artifact of a single data partition.

Because a small cross-view distance can also result from an invariant predictor, we additionally inspected the frozen oracle-relative quantity defined in Section 2.5. On the lockbox split, the asset-level mean E_{geometry} was 8.07×10^{-4} for PROP-DIRECT and 9.29×10^{-4} for B-STDPROB; the corresponding test values were 6.89×10^{-4} and 7.96×10^{-4} . These are descriptive post-hoc summaries, not a replacement for the preregistered consistency endpoint; the asset-bootstrap comparison for the information-loss regimes is reported later in Table 3.

Figure 4 visualizes the same effect sizes and confidence intervals. The close alignment of the test and lockbox estimates makes the replication pattern visible without relying on a numerical comparison alone.

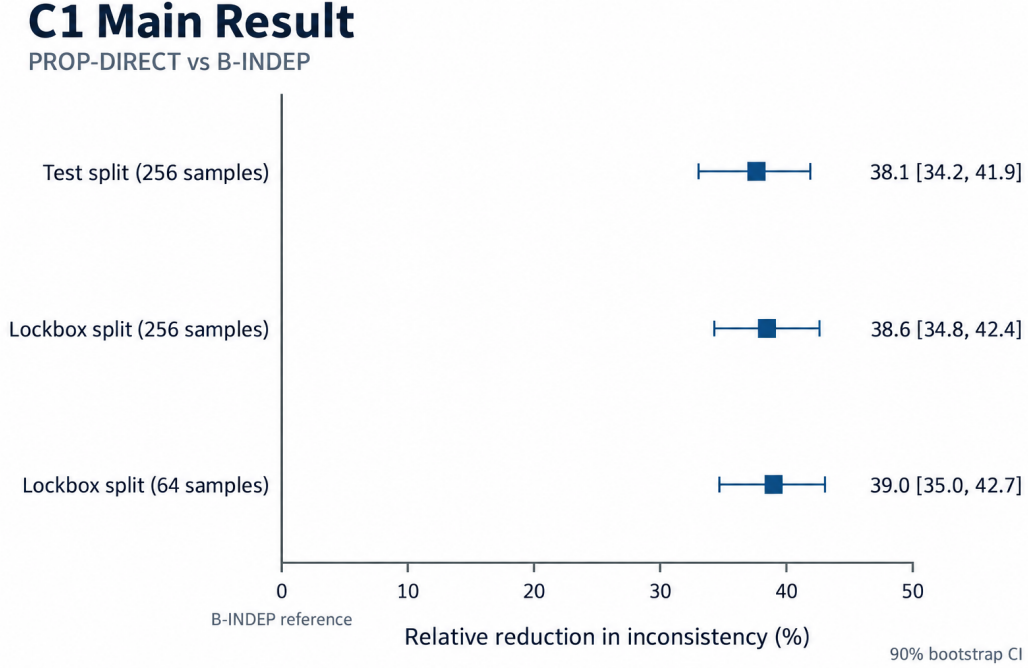


Figure 4: Primary consistency result. PROP-DIRECT reduces cross-view posterior inconsistency relative to B-INDEP by approximately 38% on both the test and lockbox splits. Squares show point estimates and horizontal bars show 90% bootstrap confidence intervals; the reduced 64-sample budget produces the same conclusion.

Predictions from different views of the same asset are therefore substantially closer in posterior geometry. The result motivates testing whether such alignment improves uncertainty-aware maintenance decisions, where prediction intervals and tail probabilities influence action.

3.1.2 Robustness to Sampling Budget

A critical deployment consideration is computational cost. Uncertainty quantification via Monte Carlo sampling typically dominates inference time, making sampling budget a key engineering constraint. We therefore evaluated whether the geometric improvement persists under reduced sampling.

When the Monte Carlo budget is reduced four-fold from 256 to 64 samples per prediction, the effect size changes by only 0.34 percentage points: 38.97% reduction (90% CI: 35.03–42.67%) versus 38.64% at the standard budget. This stability indicates that the geometric benefit does not rely on high-precision uncertainty estimation within the synthetic evaluation; whether this translates into an acceptable deployment latency trade-off requires a direct systems benchmark.

3.1.3 Interpretation

These results support the hypothesis that explicit observation-process conditioning enables models to align posterior geometry across compatible views. The architecture appears to learn that different sensor configurations, when task-sufficient, should produce posteriors that overlap appropriately rather than diverging arbitrarily. The robustness to sampling budget suggests this is a structural property of the learned representations, not a fragile artifact of estimation precision.

3.2 Boundary Behavior Under Critical Sensor Loss

The boundary comparison defined in Section 2.4 holds the number of deleted sensors fixed while changing their functional role. It asks whether the conditioning pathway distinguishes a critical deletion from a matched random deletion, rather than merely responding to the amount of missing data.

3.2.1 The Importance Boundary

PROP-DIRECT does *not* outperform the baseline in this regime. Across three pre-registered estimands measuring different aspects of geometric inconsistency under information loss, all comparisons favor the baseline or show no advantage:

- **Critical-sensor excess risk:** $\Delta R_{\text{crit}} = +8.65 \times 10^{-4}$ (90% CI: $7.36\text{--}9.91 \times 10^{-4}$)
- **Loss-geometry differential:** $\Delta G_{\text{loss}} = +1.88 \times 10^{-4}$ (90% CI: -3.55 to $+6.99 \times 10^{-4}$)
- **Interaction effect:** $I_R = +7.68 \times 10^{-4}$ (90% CI: $5.16\text{--}10.28 \times 10^{-4}$)

All point estimates are positive, indicating PROP-DIRECT exhibits *greater* inconsistency than the baseline under importance-dependent information loss. The confidence intervals either exclude zero or barely overlap it, providing consistent evidence against the superiority hypothesis.

3.2.2 Predictive-quality guardrail

The geometric degradation is accompanied by only a small change in predictive quality. A pre-specified guardrail based on continuous ranked probability score (CRPS)—a proper scoring rule for probabilistic forecasts—gives: $\Delta \text{CRPS} = +4.07 \times 10^{-4}$ (90% CI: -0.22 to $+8.47 \times 10^{-4}$), well below the non-inferiority margin of 3.85×10^{-3} .

The two metrics therefore capture different aspects of the model: the limitation concerns *geometric consistency*, while single-view RUL prediction remains competitive.

3.2.3 Pair-Class Decomposition

Examining the three comparison types reveals structure in the failure:

Figure 5 provides a visual summary of the pair-class ordering before the numerical decomposition below.

C2 Pair-Class Decomposition

PROP-DIRECT – B-STDPROB

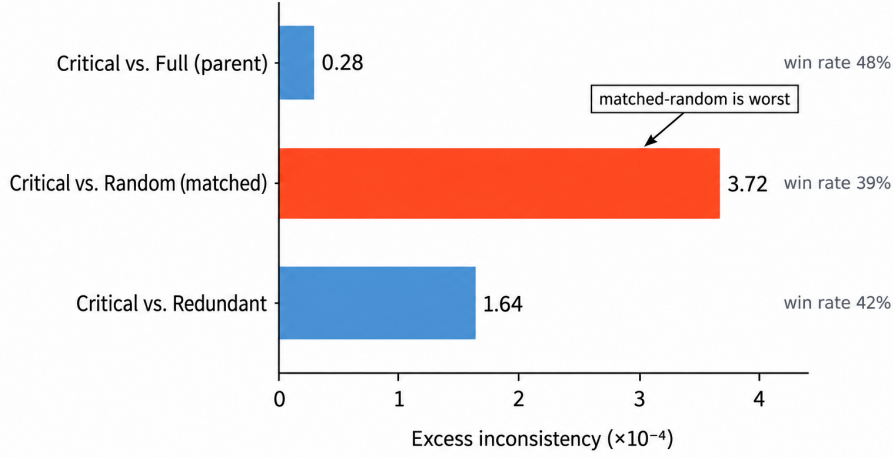


Figure 5: Information-loss pair-class decomposition. Excess inconsistency relative to B-STDPROB is small for critical-versus-full comparisons but substantially larger for the matched-random class. The associated win rates show the same ordering, highlighting the pair class in which the current flat sensor-set representation is least effective.

Comparison	Mean Difference	PROP-DIRECT Win Rate
Critical vs. Full (parent)	$+2.8 \times 10^{-5}$	48%
Critical vs. Random (matched)	$+3.72 \times 10^{-4}$	39%
Critical vs. Redundant	$+1.64 \times 10^{-4}$	42%

The pair-class ordering is informative: PROP-DIRECT is nearly at parity with the baseline for critical deletion versus the full sensor complement ($+2.8 \times 10^{-5}$ mean difference; 48% win rate), whereas its excess inconsistency is larger for random and redundant deletions. The matched-random comparison carries the largest excess inconsistency and the lowest win rate, making it the clearest stress test of whether sensor identity, rather than sensor count alone, is represented.

Figure 5 makes this asymmetry explicit: the matched-random comparison has the largest excess inconsistency and the lowest PROP-DIRECT win rate, supporting the interpretation that the architecture detects missing information more readily than functional sensor importance.

Figure 6 makes this contrast concrete for one asset at one query time: removing ancillary sensors leaves the posterior nearly unchanged, whereas removing a critical sensor widens and shifts the distribution.

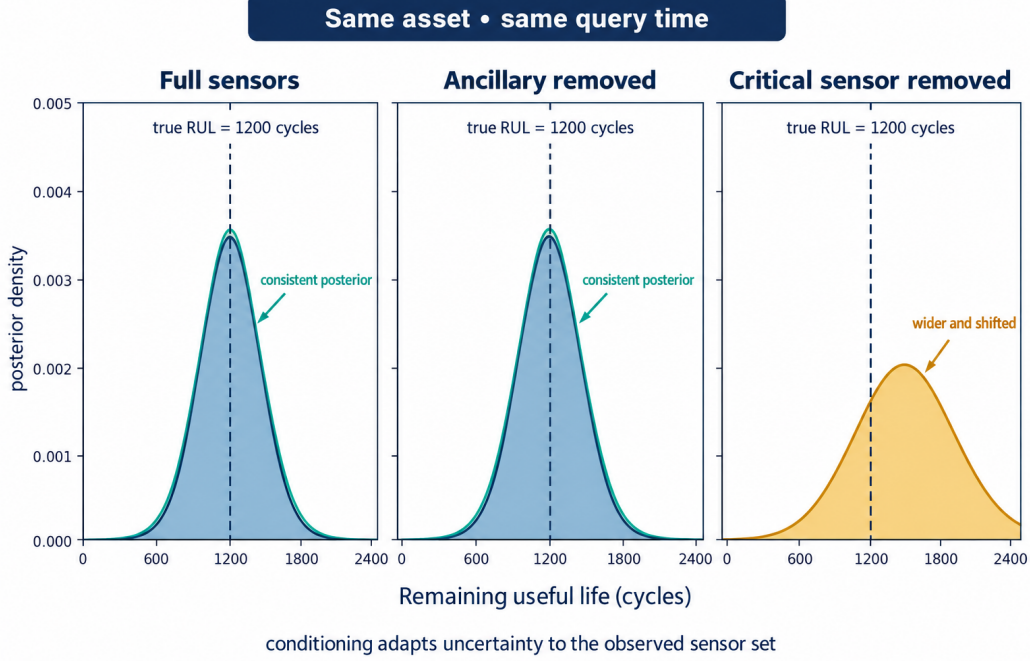


Figure 6: Illustrative prediction under changing sensor sets. For the same asset and query time, full sensors and ancillary-sensor removal produce closely overlapping posteriors, while critical-sensor removal yields a wider and shifted distribution. The example gives an operational interpretation of cross-view consistency and its importance-dependent boundary.

3.3 Exploratory Mechanistic Analysis

3.3.1 Motivation and Scope

The boundary-regime results suggest PROP-DIRECT can detect that information has changed (near-parity vs. full sensor complement) but struggles when functional importance matters (poor performance vs. matched random deletion). To investigate this mechanism, we conducted a post-hoc exploratory analysis examining how model predictions track oracle-defined task difficulty. This analysis was *not* part of the pre-registered decision criteria and should be interpreted as hypothesis-generating rather than confirmatory.

3.3.2 Oracle-relative geometry

A low cross-view distance alone would also be achieved by a model that ignores the sensor set and emits the same posterior for every view. We therefore use the frozen oracle outputs to compute the secondary quantity $E_{\text{geometry}} = |d_{\text{model}} - d_{\text{oracle}}|$ for every paired case. Table 3 reports the asset-level summary for the three deletion pair classes.

The PROP-DIRECT and B-STDPROB errors are nearly indistinguishable for the critical-versus-full comparison. PROP-DIRECT has a numerically larger residual for the matched-random and redundant comparisons, but all paired 90% intervals include zero. This secondary analysis therefore makes the correctness question explicit without turning it into a new superiority claim: the primary distance result describes consistency, while E_{geometry} describes agreement with the oracle-defined geometry.

Table 3: Exploratory oracle-relative geometry error on the C2 evaluation. Values are asset-level means of $E_{\text{geometry}} = |d_{\text{model}} - d_{\text{oracle}}|$; lower values indicate closer agreement with the oracle geometry. The final column reports the paired PROP-DIRECT minus B-STDPROB difference with a 90% asset-bootstrap interval (2,000 replicates). This analysis is secondary and was not used in any preregistered decision rule.

Pair class	PROP-DIRECT	B-STDPROB	Difference [90% CI]
Critical vs. full	0.0129	0.0129	$+2.8 \times 10^{-5}$ $[-4.3 \times 10^{-4}, +4.9 \times 10^{-4}]$
Critical vs. matched random	0.0134	0.0131	$+3.7 \times 10^{-4}$ $[-4.3 \times 10^{-4}, +1.1 \times 10^{-3}]$
Critical vs. redundant	0.0129	0.0127	$+1.6 \times 10^{-4}$ $[-3.2 \times 10^{-4}, +6.2 \times 10^{-4}]$

3.3.3 Oracle-Model Error Association

For each prediction, we retained the oracle distance and the corresponding oracle-relative geometry error. Figure 7 displays their association for PROP-DIRECT across the three comparison types. Pearson correlations are:

- **Critical vs. Full:** $r = 0.5933$ ($n = 119,232$)
- **Critical vs. Redundant:** $r = 0.5836$ ($n = 119,232$)
- **Critical vs. Random:** $r = 0.5345$ ($n = 19,872$)

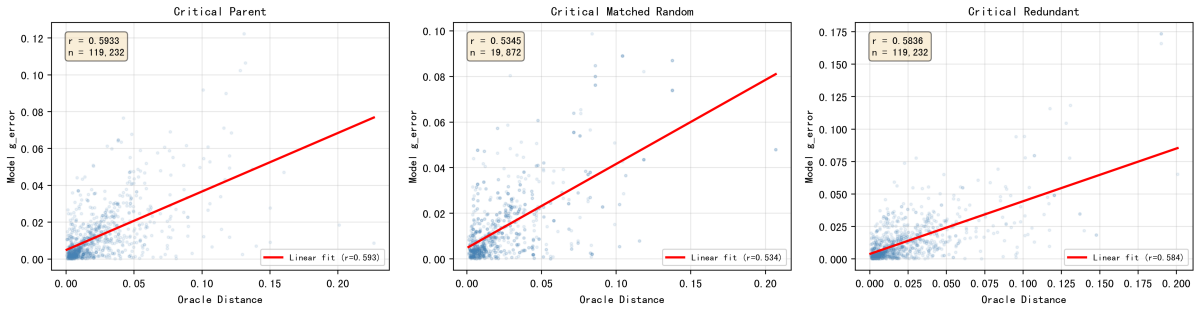


Figure 7: Oracle-relative geometry error across deletion pair classes. Each panel shows the association between oracle distance and $E_{\text{geometry}} = |d_{\text{model}} - d_{\text{oracle}}|$ for PROP-DIRECT across 1,000 randomly sampled test cases. Left: Critical vs. full sensor complement ($r = 0.5933$). Center: Critical vs. matched random deletion ($r = 0.5345$). Right: Critical vs. redundant deletion ($r = 0.5836$). The correlations are descriptive because cases share assets and query structure; they do not establish a new confirmatory endpoint.

The critical-vs-random association is 9.9% weaker than the critical-vs-full association. We treat this contrast as an exploratory pattern in oracle-relative error; the paired cases share assets and query structure.

3.3.4 Baseline Comparison: Architecture-Specific Limitation

The standard baseline (B-STDPROB) exhibits the opposite pattern:

- **Critical vs. Full:** $r = 0.6606$
- **Critical vs. Random:** $r = 0.6732$

Table 4: Descriptive predictive quality under information loss (covering the full 96-asset population; distinct from the 10-asset test/lockbox partition described in Section 2.4). Values are means within asset followed by means over 96 assets; lower is better.

Model	State MAE	Failure MAE	CRPS	State rank	Failure rank	CRPS rank
PROP-DIRECT	0.014577	0.069585	0.048445	3	3	2
B-STDPROB	0.014462	0.069195	0.048038	2	1	1
B-INDEP	0.014881	0.070197	0.049115	5	5	5
B-INVAR	0.014847	0.069970	0.048645	4	4	3
B-CDE	0.015563	0.070610	0.049103	6	6	4
B-SET	0.017344	0.071081	0.049566	7	7	7
B-WIDEN	0.014455	0.069255	0.049189	1	2	6

For B-STDPROB, the matched-random correlation is stronger. The contrast associates the weaker-correlation pattern with PROP-DIRECT’s sensor-set conditioning pathway rather than with prognostic models in general.

3.3.5 Interpretation

The residual analysis suggests that PROP-DIRECT approximates the oracle-defined geometry about as closely as the baseline in the critical-versus-full comparison, with a larger point estimate under matched-random deletion. Together with the pair-class ordering, this indicates that the availability signal is easier to represent than the functional distinctions among sensors. The evidence is diagnostic rather than a proof that either model has the correct posterior geometry in every case.

3.4 Prediction Quality and Calibration

3.4.1 Predictive quality and posterior geometry

We compared standard prognostic quality metrics across all seven model variants in the frozen model suite. Table 4 reports asset-level macro-averages on the information-loss evaluation, which is scored under a separate frozen contract covering the full 96-asset population (Section 2.4); it is distinct from the 10-asset test/lockbox partition used for the sufficient-observation claims.

PROP-DIRECT ranks third for both state mean absolute error (MAE) and failure-time MAE, and second for CRPS among the seven models. Its CRPS difference from B-STDPROB is $+4.07 \times 10^{-4}$.

The quality table separates predictive accuracy from cross-view geometry: PROP-DIRECT remains competitive on single-view metrics even where its geometric advantage disappears. The C-MAPSS analysis next asks whether this distinction is visible on a public simulated benchmark.

3.4.2 Posterior Calibration

We next use the frozen C1 posterior samples for a descriptive calibration check. The central intervals are close to their nominal levels on both held-out splits: the mean 90% coverage across the three coordinates is 0.909 for PROP-DIRECT on test and 0.902 on lockbox, compared with 0.913 and 0.903 for B-STDPROB. The PIT histograms are likewise close to the uniform reference (Figure 8); the small differences do not support a claim that either method is uniformly better calibrated. This result separates the paper’s primary geometric finding from posterior reliability:

Table 5: Posterior calibration diagnostics on the C1 evaluation splits. Coverage entries are averaged over the three joint coordinates and include 90% asset-bootstrap intervals. PIT histogram L1 is a descriptive distance from a uniform histogram; zero is ideal.

Model	Split	90% coverage	PIT histogram L1
PROP-DIRECT	test	0.909 [0.888, 0.928]	0.082
PROP-DIRECT	lockbox	0.902 [0.877, 0.925]	0.084
B-STDPROB	test	0.913 [0.894, 0.931]	0.084
B-STDPROB	lockbox	0.903 [0.877, 0.927]	0.085

the conditioning mechanism improves cross-view geometry under sufficient observation while calibration remains broadly comparable to the baseline.

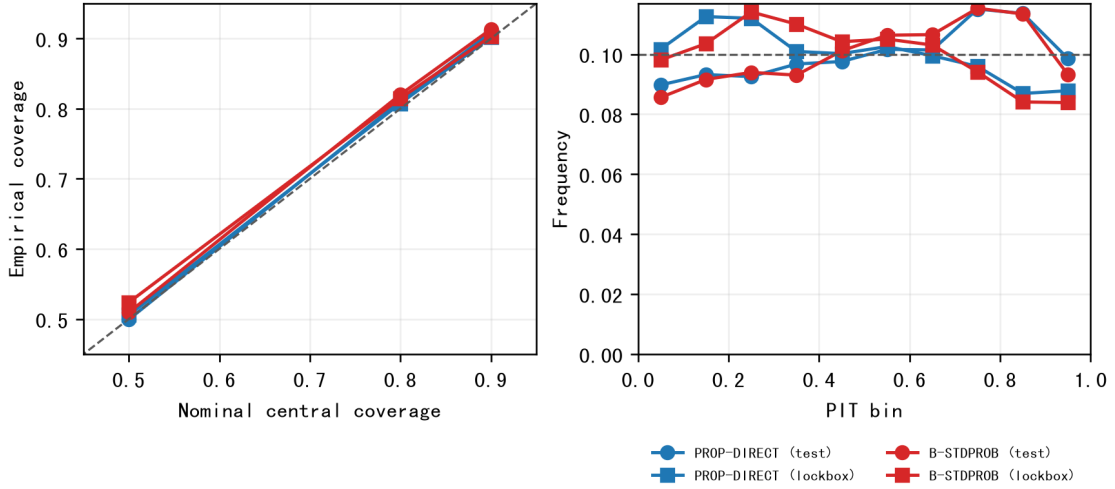


Figure 8: Posterior calibration diagnostics. Left: empirical central coverage at nominal 50%, 80%, and 90% levels, averaged over the three joint coordinates. Right: PIT histograms with the uniform reference shown by the dashed line. Circles and squares denote test and lockbox splits; the plots are descriptive checks rather than additional decision gates.

3.5 Conditional Cross-View Geometry on C-MAPSS

Using the shared-model protocol in Section 2.6, we evaluated PROP-DIRECT and B-STDPROB on C-MAPSS FD001 [32]. The benchmark provides a public simulated reference for the geometry analysis; its protocol and scope are defined before the results below.

3.5.1 Agreement depends on the view pair

Table 6 shows a conditional pattern. PROP-DIRECT reduces the distance to the full view by 18.5% (90% CI: 7.8–29.3%) for View 1 and by 28.9% (18.4–37.9%) for View 2. For the two reduced views, however, the distance ordering reverses: PROP-DIRECT exhibits a 36.6% larger cross-view distance than B-STDPROB (90% CI: –62.9 to –13.1%). The shared model therefore produces closer predictive distributions when a reduced view is compared with the full sensor complement, while the two equally sized reduced views remain farther apart under PROP-DIRECT. Without an oracle posterior, this distance ordering cannot identify which method has the more correct geometry.

Table 6: Cross-view energy distances on the C-MAPSS FD001 simulated turbofan benchmark under the shared-model protocol. A single model per architecture is trained on data that mixes all three observation configurations, mirroring the synthetic protocol. Lower energy distance indicates closer agreement between dropout-sampled predictive distributions for the same engine viewed under different sensor configurations; distances are on the raw RUL scale (cycles). Bootstrap 90% confidence intervals in brackets (2,000 replicates, resampling engines). A negative relative change means that PROP-DIRECT has a larger cross-view distance; without an oracle posterior, this is not a correctness ranking.

View Pair	PROP-DIRECT	B-STDPROB	Relative change in distance
Full vs. View 1	1.85 [1.61, 2.11]	2.27 [2.04, 2.51]	18.5% [7.8%, 29.3%]
Full vs. View 2	1.80 [1.56, 2.05]	2.54 [2.29, 2.79]	28.9% [18.4%, 37.9%]
View 1 vs. View 2	1.85 [1.57, 2.17]	1.36 [1.19, 1.52]	−36.6% [−62.9%, −13.1%]

Single-view test MAE (cycles), same runs: PROP-DIRECT 16.44 (full) / 15.95 (View 1) / 16.25 (View 2); B-STDPROB 16.27 (full) / 15.42 (View 1) / 15.73 (View 2). The near-identical point accuracy makes this geometry comparison quality-controlled: differences in cross-view distance cannot be attributed to differences in posterior informativeness.

The single-view MAEs are close across methods. PROP-DIRECT obtains 16.44, 15.95, and 16.25 cycles for the full, View 1, and View 2 conditions, compared with 16.27, 15.42, and 15.73 for B-STDPROB. The reversal in the reduced-view pair is consequently not explained by a large point-accuracy collapse. It is consistent with a more specific sensitivity to which sensors are present, although the benchmark has no oracle posterior with which to identify the mechanism or rank the two geometries by correctness.

This shared-model result is stronger evidence than the earlier specialist comparison because the conditioning pathway is trained across views, but it remains evidence from a simulated benchmark with artificial deletion. It provides a conditional cross-view result that echoes the importance-awareness question raised by the synthetic study; it does not confirm the synthetic boundary or establish performance under naturally occurring sensor loss. The archived protocol and outputs make a future study with domain-informed critical channels and field data possible.

4 Discussion

4.1 What the Results Show

The central result is a geometric one. When the available sensor views remain task-sufficient, explicitly conditioning the predictor on the observation process reduces disagreement between posterior predictions for the same asset. The 38.6% reduction in cross-view energy distance is stable across the independent test and lockbox partitions and remains after a four-fold reduction in Monte Carlo samples. The effect therefore captures a property of the representation rather than a particular sampling budget.

The oracle-relative audit supplies an important qualification. In the sufficient-view C1 scores, the descriptive residual is smaller for PROP-DIRECT than for B-STDPROB on both held-out splits, while the asset-bootstrap comparison in the information-loss regimes does not separate the two methods. The primary result should therefore be read as evidence of cross-view consistency under the stated protocol, not as a claim that a small distance alone establishes posterior correctness.

The boundary experiment gives this result its meaning. Once functionally critical sensors are removed, PROP-DIRECT no longer preserves the same advantage. Its oracle-model correlation is 9.9% weaker in the matched-random comparison, whereas B-STDPROB shows the opposite

ordering. The cases are exploratory and share assets, so the correlation is best read as a mechanistic clue. The clue is nevertheless specific: the architecture responds to the amount of available information more readily than to the role played by an individual sensor. A model can therefore be geometrically stable across views while still being insufficiently sensitive to what has been lost.

This distinction separates two questions that are often combined in missing-sensor studies. The first asks whether predictions for one asset remain compatible as its observation set changes. The second asks whether the model recognizes that different missing sensors can have different consequences. The present architecture addresses the first question under task sufficiency and exposes the second as its current boundary.

4.2 An Architectural Interpretation

PROP-DIRECT supplies the model with a representation of the available sensor set alongside the observed sequence. This gives the predictor a direct route to adjust its posterior geometry when the observation process changes. It also creates a demand on that representation: the set description must preserve distinctions that matter for degradation. The matched-random result suggests that the current flat representation does not reliably preserve those distinctions when sensor importance is strongly heterogeneous.

The baseline provides a useful comparison because it has no explicit sensor-set input. Its stronger matched-random correlation is consistent with a more conservative response to missing observations: broad uncertainty may track the severity of a critical deletion better than a configuration-specific adjustment that lacks an importance signal. This explanation is plausible rather than identified by the present experiments. It motivates a design principle for the next model generation: condition on availability, but give the conditioning pathway a structured account of sensor function.

Hierarchical representations could separate subsystem status from individual sensor status; state-dependent attention could allow importance to change with degradation mode; and physics-informed priors could encode known functional roles. These are extensions of the same geometric objective, rather than replacements for it. They should be evaluated with both sufficient-view consistency tests and deletion tests that hold sensor count fixed while changing functional importance.

4.3 Relation to Existing Prognostic Practice

The closest recent work treats missing sensing as a problem of input recovery or representation robustness. RobustUQ combines missing-value distribution modeling with conformal prediction to target valid RUL intervals [18]; MSCCAN addresses variable-sensor inputs through cross-channel attention and augmentation [14]; and GF-GGAT uses graph structure to fuse partially missing multi-sensor features [15]. Dynamic imputation and adaptive model selection provide another route to missing-data robustness [12], while Liu et al. [13] jointly optimize reconstruction and RUL prediction in a spatio-temporal multi-task network. Explicit sensor-degradation models account for changing measurement quality, from earlier state-space formulations [25, 26] to recent bearing-focused work [27]. These methods are complementary to PROP-DIRECT. Their principal outcomes are prediction error, interval validity, recovery quality, or sensor-state estimation; our primary outcome is the relation between two posterior predictions for the same asset under different observation processes.

This distinction places the C2 result in a constructive context. The present sensor-set encoder is

deliberately flat: it records which channels are available without encoding a graph of subsystem or functional relationships. The matched-random failure therefore points toward structured conditioning rather than away from the observation-process formulation. Graph-based representations such as GF-GGAT [15], hierarchical sensor groups, or state-dependent attention could supply the missing importance signal. They should be evaluated with both single-view accuracy and the paired cross-view and matched-random tests used here.

The uncertainty interpretation is also complementary to established RUL work. Sankararaman [16] emphasizes uncertainty propagation in prognostics, and Bayesian RUL frameworks distinguish epistemic from aleatoric uncertainty [17]. Recent uncertainty-aware RUL work combines Bayesian predictive distributions with explicit calibration and conformal interval construction [19]. We add an observation-process perspective: even when marginal predictive uncertainty is well calibrated, changing the sensor set can alter the relationship among posterior distributions. The calibration result and the geometry result should therefore be reported together rather than treated as interchangeable evidence of probabilistic reliability.

Set encoders and attention mechanisms [5, 6, 7] remain natural implementation alternatives, and domain-generalization methods [23, 24] offer useful training strategies for operating-condition shifts. Their inclusion does not by itself establish importance-awareness; the paired estimands in this study provide a direct test of whether a model distinguishes a change in sensor count from a change in sensor role.

4.4 The C-MAPSS Result in Context

The synthetic generator gives the primary experiment a property that field data cannot easily provide: an exact oracle posterior and direct control over which observation process is applied to the same latent asset. C-MAPSS FD001 offers a complementary public reference point, but it is also a simulated benchmark. In the shared-model evaluation, one predictor per method was trained on all three views, the predictors used an MSE point head with dropout sampling, and each reduced view was formed by artificial deletion of three of the 14 active channels.

The result is conditional. PROP-DIRECT has 18.5–28.9% smaller distance to the full view, but a 36.6% larger distance than B-STDPROB for the pair of equally sized reduced views. Single-view MAE is close across methods, so the reversal does not track a broad point-accuracy collapse. Because C-MAPSS has no oracle posterior for these artificial views, the larger distance cannot be interpreted as a less correct posterior; it is evidence that the two methods respond differently to sensor identity. The result is protocol-aligned evidence from a simulated benchmark, not field validation; data with naturally occurring sensor loss would test whether the same boundary matters in operation.

4.5 Scope and Future Tests

Several aspects of the evidence define the next experiments rather than weaken the result already established. The synthetic study uses one generator family and focuses on RUL posterior geometry; the calibration diagnostic is broadly comparable between methods, but replication with independent generators, tail-risk metrics, and other prognostic tasks would test the breadth of the finding. The present model comparison also leaves attention-based, ensemble, and importance-aware architectures open. These comparisons are especially valuable because a model that is accurate in a single view can still be poorly aligned across views, while a model that is highly aligned can achieve that agreement by becoming insufficiently informative.

For practical evaluation, the most informative sequence is therefore straightforward: first train

a shared model across observation configurations on the public benchmark, then repeat the deletion study with domain-informed critical channels, and finally examine longitudinal assets with naturally occurring outages. If these tests preserve the geometry gain while improving the importance boundary, observation-process conditioning would offer a useful component for uncertainty-aware maintenance decisions. If they do not, the present study still identifies the relevant failure mode and the metric pair needed to detect it.

5 Conclusion

Reliable prognostics must remain coherent when the sensors available to an asset change. In a controlled synthetic evaluation with exact oracle posteriors, explicitly conditioning the predictor on the observed sensor set reduced cross-view posterior inconsistency by 38.6% (90% CI: 34.8–42.4%) when the alternative views remained task-sufficient. The effect persisted after a four-fold reduction in Monte Carlo sampling, suggesting that the gain is tied to the representation of the observation process rather than to a large sampling budget.

The same study identifies the boundary of that gain. Under permanent removal of functionally critical sensors, PROP-DIRECT did not retain its advantage, while exploratory oracle-model correlations showed a weaker distinction between critical and random deletions than the standard baseline. These comparisons are descriptive because the cases share assets and query structure, but they point to a concrete architectural issue: recording which sensors are present is easier than representing what those sensors mean for the degradation process. The result is a useful design constraint for future importance-aware conditioning, rather than a reason to discard observation-process conditioning altogether.

On C-MAPSS FD001, the shared-model evaluation reduces distance to the full view by 18.5–28.9%, but produces a 36.6% larger distance than the baseline between the two equally sized reduced views. Point MAE remains close to the baseline. Because C-MAPSS supplies no oracle posterior for the artificial views, this result shows a conditional difference in cross-view behavior rather than a correctness ranking; evaluation with naturally occurring sensor loss remains the next test.

Taken together, the findings support a simple principle for probabilistic prognostics: observation-process conditioning is valuable when available views preserve the information needed by the task, and it needs an explicit account of functional importance when they do not. Hierarchical sensor representations, state-dependent importance weights, and physics-informed priors offer natural ways to pursue that account while retaining the geometric objective introduced here.

Data Availability Statement

The public code and manuscript repository is available at <https://github.com/xyforge/sensor-set-conditioning-rul>. It contains the manuscript source, synthetic-generator specifications and seed settings, evaluation scripts, shared-model C-MAPSS configurations and results, calibration summaries, compact score artifacts, and the frozen protocol documents. All evaluation code and decision rules were frozen before lockbox validation.

Raw C-MAPSS data, large prediction stores, and model checkpoints are excluded from the repository and can be obtained or regenerated under their respective access conditions. A persistent DOI can be added when a versioned release is deposited in a research repository.

Acknowledgements

This research was conducted under a preregistered protocol with independent test and lockbox evaluation splits. C-MAPSS turbofan data were provided by NASA Ames Prognostics Center of Excellence. Computational resources were provided by Shanghai University of Engineering Science. All experimental decisions were made prior to accessing validation outcomes to ensure statistical integrity.

Conflict of Interest Statement

The authors declare no competing financial interests or personal relationships that could influence the work reported in this paper.

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